

# GearDesign Report

Cylindrical gear(pair)

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# 1. Summary

## 1.1. Gear geometry

Meshing pair: Gear1 - Gear2

|                                              | Symbol     | Pair1<br>Gear1                             | Pair1<br>Gear2                             |
|----------------------------------------------|------------|--------------------------------------------|--------------------------------------------|
| Normal module (mm)                           | $m_n$      | 3.0000                                     |                                            |
| Pressure angle at normal section (deg)       | $\alpha_n$ | 20.0000                                    |                                            |
| Helix angle at reference circle (deg)        | $\beta$    | 0.0000                                     |                                            |
| Manual normal backlash (mm)                  | $j_{bn}$   | 0.0000                                     |                                            |
| Number of teeth                              | $z$        | 28.0000                                    | 41.0000                                    |
| Facewidth (mm)                               | $b$        | 45.0000                                    | 45.0000                                    |
| Profile shift coefficient                    | $x$        | 0.3132                                     | 0.2129                                     |
| Quality (ISO)                                | $Q$        | 6.0000                                     | 6.0000                                     |
| Center distance (mm)                         | $a$        | 105.0000                                   |                                            |
| Tooth thickness tolerance                    | -          | DIN 3967 cd25                              | DIN 3967 cd25                              |
| Gear material                                | -          | DIN 18CrNiMo7 ( $\sigma_{Hlim} = 430$ Mpa) | DIN 18CrNiMo7 ( $\sigma_{Hlim} = 430$ Mpa) |
| Reference diameter (mm)                      | $d$        | 84.0000                                    | 123.0000                                   |
| Tip diameter (mm)                            | $d_a$      | 91.879 (91.879 / 91.779)                   | 130.277 (130.277 / 130.177)                |
| Root diameter (mm)                           | $d_f$      | 78.379 (78.187 / 78.077)                   | 116.777 (116.585 / 116.475)                |
| Number of teeth spanned                      | $k$        | 4.0000                                     | 5.0000                                     |
| Base tangent length (no backlash) (mm)       | $W_k$      | 32.817 (32.751 / 32.713)                   | 42.013 (41.947 / 41.910)                   |
| Effective Diameter of ball/pin (mm)          | $D_M$      | 5.5000                                     | 5.2500                                     |
| Diametral measurement over two balls<br>(mm) | $M_{dk}$   | 93.799 (93.652 / 93.567)                   | 131.519 (131.356 / 131.262)                |

## 1.2. Total safety factor

|                               | Gear1<br>[Left] | Gear1<br>[Right] | Gear2<br>[Left] | Gear2<br>[Right] | Required<br>S.F |
|-------------------------------|-----------------|------------------|-----------------|------------------|-----------------|
| Contact S.F                   | 0.7023          |                  | 0.7246          |                  | 1.0000          |
| Bending S.F                   | 0.8932          |                  | 0.8829          |                  | 1.4000          |
| Scuffing S.F<br>(Flash Temp.) | 3.6689          |                  |                 |                  | 2.0000          |
| Micropitting S.F              | 0.4053          |                  |                 |                  | 1.0000          |

## 1.3. Total damage & life

|                      | Gear1<br>[Left] | Gear1<br>[Right] | Gear2<br>[Left] | Gear2<br>[Right] |
|----------------------|-----------------|------------------|-----------------|------------------|
| Contact Damage [%]   | 9999.9000       | 0.0000           | 9999.9000       | 0.0000           |
| Bending Damage [%]   | 9999.9000       | 0.0000           | 9999.9000       | 0.0000           |
| Contact Life [hr]    | 43.9000         | $\infty$         | 79.1000         | $\infty$         |
| Bending Life [hr]    | 3.1000          | $\infty$         | 3.7000          | $\infty$         |
| Total Req. Life [hr] | 10000.0000      |                  |                 |                  |

## 1.4. Others

|                                           | Gear1 - Gear2 |
|-------------------------------------------|---------------|
| Transverse Contact Ratio                  | 1.5335        |
| Overlap Ratio                             | 0.0000        |
| Total Contact Ratio                       | 1.5335        |
| Worst Pitch Line Velocity [m/s]           | 4.4620        |
| Total Efficiency [%]                      | 99.1509       |
| Worst Mesh Loss [W]                       | 863.2802      |
| Worst Churning Loss [W]                   | 47.5358       |
| Worst Peak-to-Peak T.E. [ $\mu\text{m}$ ] | 15.7414       |
| Gear Total Mass [kg]                      | 2.6096        |
| Hunting tooth                             | Complete      |

## 1.5. Summary - Load cases

[illegible]

## 2. Detail geometry

### 2.1. Input gear geometry

Meshing pair: Gear1 - Gear2

|                                                    | Symbol           | Pair1<br>Gear1                             | Pair1<br>Gear2                             |
|----------------------------------------------------|------------------|--------------------------------------------|--------------------------------------------|
| Normal module (mm)                                 | $m_n$            | 3.0000                                     |                                            |
| Pressure angle at normal section (deg)             | $\alpha_n$       | 20.0000                                    |                                            |
| Helix angle at reference circle (deg)              | $\beta$          | 0.0000                                     |                                            |
| Manual normal backlash (mm)                        | $j_{bn}$         | 0.0000                                     |                                            |
| Number of teeth                                    | $z$              | 28.0000                                    | 41.0000                                    |
| Facewidth (mm)                                     | $b$              | 45.0000                                    | 45.0000                                    |
| Profile shift coefficient                          | $x$              | 0.3132                                     | 0.2129                                     |
| Quality (ISO)                                      | $Q$              | 6.0000                                     | 6.0000                                     |
| Center distance (mm)                               | $a$              | 105.0000                                   |                                            |
| Tooth thickness tolerance                          | -                | DIN 3967 cd25                              | DIN 3967 cd25                              |
| Addendum coefficient                               | $h^*_aP$         | 1.0000                                     | 1.0000                                     |
| Dedendum coefficient                               | $h^*_fP$         | 1.2500                                     | 1.2500                                     |
| Root radius coefficient                            | $\rho^*_fP$      | 0.3800                                     | 0.3800                                     |
| Tip form height coefficient                        | $h^*_{FaP}$      | 0.9667                                     | 0.9667                                     |
| Ramp angle (deg)                                   | $\alpha_{KP}$    | 50.0000                                    | 50.0000                                    |
| Protuberance height coefficient                    | $h^*_{prP}$      | 0.0000                                     | 0.0000                                     |
| Protuberance angle (deg)                           | $\alpha_{prP}$   | 0.0000                                     | 0.0000                                     |
| Tip alteration (mm)                                | $k_{m_n}$        | 0.0000                                     | 0.0000                                     |
| Tip radius coefficient (for topping)               | $\rho^*_{aP}$    | 0.0000                                     | 0.0000                                     |
| Gear material                                      | -                | DIN 18CrNiMo7 ( $\sigma_{Hlim} = 430$ Mpa) | DIN 18CrNiMo7 ( $\sigma_{Hlim} = 430$ Mpa) |
| Young's modulus (N/mm <sup>2</sup> )               | $E$              | $2.0600 \times 10^5$                       | $2.0600 \times 10^5$                       |
| Poisson's ratio                                    | $\nu$            | 0.3000                                     | 0.3000                                     |
| Density (kg/m <sup>3</sup> )                       | $\rho$           | 7830.0000                                  | 7830.0000                                  |
| Coeff. of thermal expansion (10 <sup>-8</sup> /°C) | $\alpha_{coeff}$ | 11.5000                                    | 11.5000                                    |
| Surface hardness                                   | HRC              | 61.0000                                    | 61.0000                                    |
| Core hardness                                      | HBW              | 325.0000                                   | 325.0000                                   |
| Tensile strength (N/mm <sup>2</sup> )              | $\sigma_B$       | 1200.0000                                  | 1200.0000                                  |
| Yield strength (N/mm <sup>2</sup> )                | $\sigma_s$       | 850.0000                                   | 850.0000                                   |
| Specific heat capacity (J/kg.K)                    | $c_M$            | 485.0000                                   | 485.0000                                   |
| Specific heat conductivity (W/m.K)                 | $\lambda_M$      | 50.0000                                    | 50.0000                                    |
| Mean roughness height, root (um)                   | $R_{ZF}$         | 20.0000                                    | 20.0000                                    |
| Mean roughness height, flank (um)                  | $R_{ZH}$         | 4.8000                                     | 4.8000                                     |
| Roughness average value, root (um)                 | $R_{AF}$         | 3.0000                                     | 3.0000                                     |
| Roughness average value, flank (um)                | $R_{AH}$         | 0.6000                                     | 0.6000                                     |

|                                                             | Symbol          | Pair1<br>Gear1 | Pair1<br>Gear2 |
|-------------------------------------------------------------|-----------------|----------------|----------------|
| Fatigue strength for bending stress<br>(N/mm <sup>2</sup> ) | $\sigma_{Flim}$ | 430.0000       | 430.0000       |
| Fatigue strength for contact stress (N/mm <sup>2</sup> )    | $\sigma_{Hlim}$ | 1500.0000      | 1500.0000      |

## 2.2. Calculated geometry - Gear

Meshing pair: Gear1 - Gear2

|                                                                           | Symbol                 | Pair1<br>Gear1    | Pair1<br>Gear2      |
|---------------------------------------------------------------------------|------------------------|-------------------|---------------------|
| Gear ratio                                                                | GR                     | 1.4643            |                     |
| Transverse module (mm)                                                    | $m_t$                  | 3.0000            |                     |
| Working normal pressure angle (deg)                                       | $\alpha_{wn}$          | 22.1396           |                     |
| Transverse pressure angle (deg)                                           | $\alpha_t$             | 20.0000           |                     |
| Working transverse pressure angle (deg)                                   | $\alpha_{wt}$          | 22.1396           |                     |
| Working transverse pressure angle, e/i (deg)                              | $\alpha_{wt}, e/i$     | 22.1543 / 22.1248 |                     |
| Base helix angle (deg)                                                    | $\beta_b$              | 0.0000            |                     |
| Helix angle at operating pitch circle (deg)                               | $\beta_w$              | 0.0000            |                     |
| Radial backlash without tolerance (mm)                                    | $j_{r0}$               | 0.0000            |                     |
| Circumferential backlash without tolerance (mm)                           | $j_{t0}$               | 0.0000            |                     |
| Min/max circumferential backlash by flank tolerance<br>for each gear (mm) | $j_t, e/i$             | 0.0700 / 0.1100   | 0.0700 / 0.1100     |
| Min/max normal backlash by flank tolerance<br>for each gear (mm)          | $j_n, e/i$             | 0.0658 / 0.1034   | 0.0658 / 0.1034     |
| Circumferential backlash, e/i (mm)                                        | $j_{tw}, e/i$          | 0.1331 / 0.2321   |                     |
| Normal backlash, e/i (mm)                                                 | $j_{nw}, e/i$          | 0.1233 / 0.2150   |                     |
| Angular backlash, min (deg)                                               | $j_\theta, \text{min}$ | 0.1789            | 0.1222              |
| Angular backlash, max (deg)                                               | $j_\theta, \text{max}$ | 0.3122            | 0.2132              |
| Effective facewidth (mm)                                                  | $b_{eff}$              | 45.0000           |                     |
| Sum of the profile shift coefficients                                     | Sum_x                  | 0.5261            |                     |
| Generating profile shift coefficient                                      | $x_E, e/i$             | 0.2811 / 0.2628   | 0.1808 / 0.1625     |
| Prevent undercut profile shift coefficient                                | $x_{Emin}$             | -0.6377           | -1.3981             |
| Max. profile shift coeff. (min top land)                                  | $x_{Emax}$             | 1.2297            | 1.6118              |
| Reference center distance, $x_1=x_2=0$ (mm)                               | $a_d$                  | 103.5000          |                     |
| Working center distance (mm)                                              | $a_w$                  | 105.0000          |                     |
| Reference diameter (mm)                                                   | $d$                    | 84.0000           | 123.0000            |
| Operating pitch diameter (mm)                                             | $d_w$                  | 85.2174           | 124.7826            |
| Operating pitch diameter with tolerance (mm)                              | $d_w, e/i$             | 85.2263 / 85.2085 | 124.7957 / 124.7695 |
| 27                                                                        |                        | 91.8792           | 130.2772            |
| Tip diameter with tolerance (mm)                                          | $d_a, e/i$             | 91.8792 / 91.7792 | 130.2772 / 130.1772 |

|                                                             | Symbol    | Pair1<br>Gear1    | Pair1<br>Gear2      |
|-------------------------------------------------------------|-----------|-------------------|---------------------|
| Root diamter (mm)                                           | d_f       | 78.3792           | 116.7772            |
| Root diameter with tolerance (mm)                           | d_f, e/i  | 78.1869 / 78.0770 | 116.5848 / 116.4749 |
| Tip form diameter (mm)                                      | d_Fa      | 91.6792           | 130.0772            |
| Tip form diameter with tolerance (mm)                       | d_Fa, e/i | 91.4572 / 91.3306 | 129.8643 / 129.7430 |
| Root form diamter (mm)                                      | d_Ff      | 80.6777           | 118.9869            |
| Root form diameter with tolerance (mm)                      | d_Ff, e/i | 80.5633 / 80.4996 | 118.8546 / 118.7802 |
| Active tip diameter (EAP) (mm)                              | d_Na      | 91.6792           | 130.0772            |
| Active tip diameter with tolerance (mm)                     | d_Na, e/i | 91.4572 / 91.3306 | 129.8643 / 129.7430 |
| Active root diameter (SAP) (mm)                             | d_Nf      | 81.2997           | 120.0673            |
| Active root diameter with tolerance (mm)                    | d_Nf, e/i | 81.4926 / 81.3981 | 120.2719 / 120.1707 |
| Mean diameter for mass calculation (mm)                     | d_m       | 84.9806           | 123.3785            |
| Gear mass (kg)                                              | m         | 1.0131            | 1.5965              |
| Gear inertia (kg.m^2)                                       | J         | 0.0000            | 0.0000              |
| Nominal tip clearance (mm)                                  | c         | 0.6718            | 0.6718              |
| Effective tip clearance, e/i (mm)                           | c, e/i    | 0.8839 / 0.7570   | 0.8839 / 0.7570     |
| Min. margin of involute contact, (d_Nf_i - d_Ff_e) / 2 (mm) | c_Nff     | 0.4174            | 0.6580              |

## 2.3. Calculated geometry - Tooth

Meshing pair: Gear1 - Gear2

|                                          | Symbol     | Pair1<br>Gear1  | Pair1<br>Gear2  |
|------------------------------------------|------------|-----------------|-----------------|
| Tooth depth (mm)                         | h          | 6.7500          | 6.7500          |
| Tooth depth, e/i (mm)                    | h, e/i     | 6.9011 / 6.7962 | 6.9011 / 6.7962 |
| Working depth (mm)                       | h_w        | 6.0782          |                 |
| Addendum (mm)                            | h_a        | 3.9396          | 3.6386          |
| Addendum, e/i (mm)                       | h_a, e/i   | 3.9396 / 3.8896 | 3.6386 / 3.5886 |
| Dedendum (mm)                            | h_f        | 2.8104          | 3.1114          |
| Dedendum, e/i (mm)                       | h_f, e/i   | 2.9066 / 2.9615 | 3.2076 / 3.2625 |
| Number of virtual gear teeth             | z_n        | 28.0000         | 41.0000         |
| Tooth thickness coeff.                   | s_nc       | 1.7988          | 1.7257          |
| Transverse tooth thickness (mm)          | s_t        | 5.3964          | 5.1772          |
| Normal tooth thickness (mm)              | s_n        | 5.3964          | 5.1772          |
| Normal tooth thickness at d_a (mm)       | s_an       | 1.9011          | 2.1504          |
| Normal tooth thickness at d_a, e/i (mm)  | s_an, e/i  | 1.8820 / 1.7808 | 2.1266 / 2.0339 |
| Normal tooth thickness at d_Fa (mm)      | s_Fan      | 2.0155          | 2.2507          |
| Normal tooth thickness at d_Fa, e/i (mm) | s_Fan, e/i | 2.1357 / 2.0212 | 2.3423 / 2.2402 |
| Normal space width at d_f (mm)           | e_fn       | 0.0000          | 2.4074          |
| Normal space width at d_f, e/i (mm)      | e_fn, e/i  | 0.0000 / 0.0000 | 2.4432 / 2.4647 |



|                            | Symbol | Pair1<br>Gear1 | Pair1<br>Gear2 |
|----------------------------|--------|----------------|----------------|
| Normal pitch (mm)          | p_n    | 9.4248         |                |
| Transverse pitch (mm)      | p_t    | 9.4248         |                |
| Normal base pitch (mm)     | p_bn   | 8.8564         |                |
| Transverse base pitch (mm) | p_bt   | 8.8564         |                |
| Axial pitch (mm)           | p_x    | 0.0000         |                |
| Lead (mm)                  | p_z    | 0.0000         | 0.0000         |

## 2.4. Calculated geometry - Mesh

Meshing pair: Gear1 - Gear2

|                                                | Symbol    | Pair1<br>Gear1              | Pair1<br>Gear2 |
|------------------------------------------------|-----------|-----------------------------|----------------|
| Length of the path of contact (A to E) (mm)    | g_α       | 13.5812                     |                |
| Length of path of contact, e/i (mm)            | g_α, e/i  | 13.1587 / 12.8415           |                |
| Length of approach POC of pinion (A to C) (mm) | g_f1      | 6.3233                      |                |
| Length of approach POC of pinion, e/i (mm)     | g_f1, e/i | 6.1080 / 6.0733             |                |
| Length of recess POC of pinion (C to E) (mm)   | g_a1      | 7.2579                      |                |
| Length of recess POC of pinion, e/i (mm)       | g_a1, e/i | 7.0508 / 7.0271             |                |
| Length T1 - A (mm)                             | T1A       | 9.7344                      |                |
| T1A tolerance (mm)                             | e/m/i     | 10.1296 / 10.0337 / 9.9379  |                |
| Length T1 - B (mm)                             | T1B       | 14.4592                     |                |
| T1B tolerance (mm)                             | e/m/i     | 14.2402 / 14.1774 / 14.1147 |                |
| Length T1 - C (mm)                             | T1C       | 16.0577                     |                |
| T1C tolerance (mm)                             | e/m/i     | 16.0695 / 16.0577 / 16.0458 |                |
| Length T1 - D (mm)                             | T1D       | 18.5908                     |                |
| T1D tolerance (mm)                             | e/m/i     | 18.9860 / 18.8901 / 18.7943 |                |
| Length T1 - E (mm)                             | T1E       | 23.3156                     |                |
| T1E tolerance (mm)                             | e/m/i     | 23.0966 / 23.0338 / 22.9711 |                |
| Length T2 - A (mm)                             | T2A       | 29.8363                     |                |
| T2A tolerance (mm)                             | e/m/i     | 29.6037 / 29.5370 / 29.4703 |                |
| Length T2 - B (mm)                             | T2B       | 25.1115                     |                |
| T2B tolerance (mm)                             | e/m/i     | 25.4852 / 25.3933 / 25.3013 |                |
| Length T2 - C (mm)                             | T2C       | 23.5130                     |                |
| T2C tolerance (mm)                             | e/m/i     | 23.5304 / 23.5130 / 23.4957 |                |
| Length T2 - D (mm)                             | T2D       | 20.9799                     |                |
| T2D tolerance (mm)                             | e/m/i     | 20.7473 / 20.6806 / 20.6139 |                |
| Length T2 - E (mm)                             | T2E       | 16.2551                     |                |
| T2E tolerance (mm)                             | e/m/i     | 16.6289 / 16.5369 / 16.4449 |                |
| Length T1 - T2 (mm)                            | T1T2      | 39.5707                     |                |

|                                         | Symbol            | Pair1<br>Gear1              | Pair1<br>Gear2 |
|-----------------------------------------|-------------------|-----------------------------|----------------|
| T1T2 tolerance (mm)                     | e/m/i             | 39.5999 / 39.5707 / 39.5415 |                |
| Diameter of single contact point A (mm) | d_A               | 81.2997                     | 130.0772       |
| Diameter of single contact point B (mm) | d_B               | 84.0647                     | 126.0222       |
| Diameter of single contact point C (mm) | d_C               | 85.2174                     | 124.7826       |
| Diameter of single contact point D (mm) | d_D               | 87.2529                     | 122.9629       |
| Diameter of single contact point E (mm) | d_E               | 91.6792                     | 120.0673       |
| Transverse contact ratio                | $\epsilon_\alpha$ | 1.5335                      |                |
| Overlap ratio                           | $\epsilon_\beta$  | 0.0000                      |                |
| Total contact ratio                     | $\epsilon_Y$      | 1.5335                      |                |
| Total contact ratio with tolerance      | e/m/i             | 1.486 / 1.468 / 1.450       |                |
| Specific sliding at Tip                 | $\zeta_a$         | 0.5239                      | 0.5223         |
| Specific sliding at Root                | $\zeta_f$         | -1.0932                     | -1.1003        |

## 2.5. Calculated geometry - Accuracy

Meshing pair: Gear1 - Gear2

|                                                             | Symbol  | Pair1<br>Gear1 | Pair1<br>Gear2 |
|-------------------------------------------------------------|---------|----------------|----------------|
| Quality Standard                                            | -       | ISO 1328: 2013 | ISO 1328: 2013 |
| Single pitch deviation ( $\mu\text{m}$ )                    | f_pt    | 9.0000         | 9.0000         |
| Base circle pitch deviation ( $\mu\text{m}$ )               | f_pb    | 8.5000         | 8.5000         |
| Sector pitch deviation over k/8 pitches ( $\mu\text{m}$ )   | F_pk/8T | 20.0000        | 19.0000        |
| Profile form deviation ( $\mu\text{m}$ )                    | f_fa    | 9.5000         | 9.5000         |
| Profile slope deviation ( $\mu\text{m}$ )                   | f_Ha    | 7.5000         | 7.5000         |
| Total profile deviation ( $\mu\text{m}$ )                   | F_α     | 12.0000        | 12.0000        |
| Helix form deviation ( $\mu\text{m}$ )                      | f_fb    | 11.0000        | 11.0000        |
| Helix slope deviation ( $\mu\text{m}$ )                     | f_Hβ    | 9.5000         | 10.0000        |
| Total helix deviation ( $\mu\text{m}$ )                     | F_β     | 15.0000        | 15.0000        |
| Cumulative pith deviation, total ( $\mu\text{m}$ )          | F_p     | 27.0000        | 29.0000        |
| Runout ( $\mu\text{m}$ )                                    | F_r     | 24.0000        | 26.0000        |
| Single flank composite, tooth-to-tooth ( $\mu\text{m}$ )    | f_isT   | 8.5000         | 8.5000         |
| Single flank composite, total ( $\mu\text{m}$ )             | F_isT   | 35.5000        | 37.5000        |
| Tooth-to-tooth radial composite deviation ( $\mu\text{m}$ ) | f_idT   | 14.0000        | 14.0000        |
| Total radial composite deviation ( $\mu\text{m}$ )          | F_idT   | 36.0000        | 36.0000        |

## 2.6. Calculated geometry - Measurement

Meshing pair: Gear1 - Gear2

|                                                    | Symbol             | Pair1<br>Gear1    | Pair1<br>Gear2      |
|----------------------------------------------------|--------------------|-------------------|---------------------|
| Number of teeth spanned                            | k                  | 4.0000            | 5.0000              |
| Base tangent length (mm)                           | W_k                | 32.8166           | 42.0133             |
| Base tangent length with tolerance (mm)            | W_k, e/i           | 32.7508 / 32.7132 | 41.9475 / 41.9099   |
| Tolerance of W_k (mm)                              | $\Delta W_k$ , e/i | -0.0658 / -0.1034 | -0.0658 / -0.1034   |
| Diameter of contact point (mm)                     | d_M                | 85.4517           | 122.9523            |
| Theoretical diameter of ball/pin (mm)              | D_M.theory         | 5.4396            | 5.2205              |
| Effective Diameter of ball/pin (mm)                | D_M.eff            | 5.5000            | 5.2500              |
| Radial single ball measurement (mm)                | M_rk               | 46.8996           | 65.8060             |
| Radial single ball measurement with tolerance (mm) | M_rk, e/i          | 46.8260 / 46.7837 | 65.7243 / 65.6774   |
| Diametral two ball measurement (mm)                | M_dk               | 93.7992           | 131.5193            |
| Diametral two ball measurement with tolerance (mm) | M_dk, e/i          | 93.6520 / 93.5674 | 131.3561 / 131.2623 |
| Tolerance of M_dk (mm)                             | $\Delta M_{dk}$    | -0.1473 / -0.2319 | -0.1632 / -0.2569   |

## 3. Detail rating

### 3.1. General

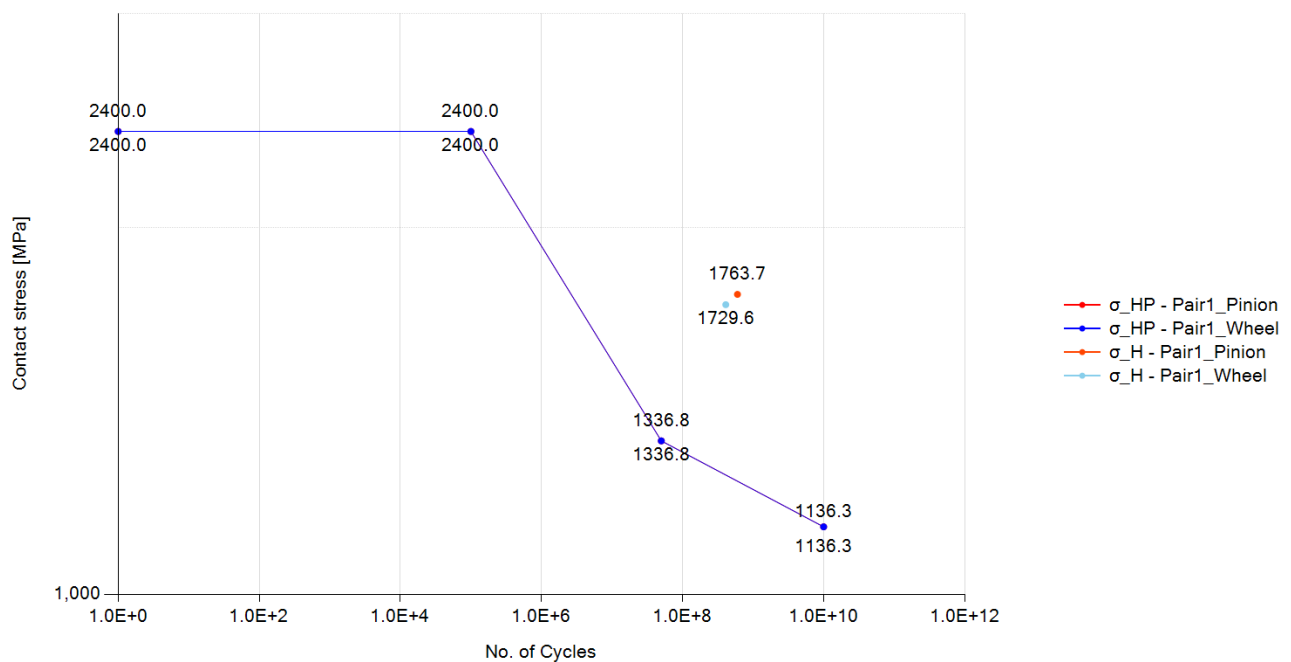
Load case 1, Meshing pair: Gear1 - Gear2

|                                                                  | Symbol        | Pair1<br>Gear1    | Pair1<br>Gear2 |
|------------------------------------------------------------------|---------------|-------------------|----------------|
| Rotational speed (rpm)                                           | $\omega$      | 1000.0000         | 682.9270       |
| Nominal tangential load at d (N)                                 | $F_t$         | 24390.2440        |                |
| Nominal axial load at d (N)                                      | $F_a$         | 0.0000            |                |
| Nominal radial load at d (N)                                     | $F_r$         | 8877.3230         |                |
| Nominal normal load at d (N)                                     | $F_n$         | 25955.5550        |                |
| Nominal tangential load per unit facewidth (N/mm)                | $w$           | 542.0050          |                |
| Circumferential speed at d (m/s)                                 | $v$           | 4.3980            |                |
| Sliding velocity at Tip dia. (m/s)                               | $v_g$         | 1.2790            | 1.1140         |
| Nominal tangential load at d_w (N)                               | $F_{tw}$      | 24041.8140        |                |
| Nominal axial load at d_w (N)                                    | $F_{aw}$      | 0.0000            |                |
| Nominal radial load at d_w (N)                                   | $F_{rw}$      | 9781.7190         |                |
| Circumferential speed at d_w (m/s)                               | $v_w$         | 4.4620            |                |
| Running-in value (um)                                            | $y_p$         | 0.6370            |                |
| Running-in value (um)                                            | $y_f$         | 0.7130            |                |
| Effective single pitch & profile deviation (um)                  | $f_{pb\_eff}$ | 7.8620            |                |
| Effective single pitch & profile deviation (um)                  | $f_{fa\_eff}$ | 8.7870            |                |
| Correction factor                                                | $C_M$         | 0.8000            |                |
| Gear blank factor                                                | $C_R$         | 0.8980            |                |
| Basic rack factor                                                | $C_B$         | 0.9750            |                |
| Material factor                                                  | $E/E_{st}$    | 1.0000            |                |
| Theoretical single stiffness (N/mm/um)                           | $c'_{th}$     | 18.2760           |                |
| Single tooth stiffness (N/mm/um)                                 | $c'$          | 12.8010           |                |
| Mesh stiffness (for $K_v$ , $KH\alpha$ , $KF\alpha$ ), (N/mm/um) | $c_{y\alpha}$ | 17.9230           |                |
| Mesh stiffness (for $KH\beta$ , $KF\beta$ ), (N/mm/um)           | $c_{y\beta}$  | 15.2350           |                |
| Mean diameter for $K_v$ (mm)                                     | $d_m(K_v)$    | 85.1290           | 123.5270       |
| Moment of inertia for $K_v$ (kg.mm <sup>2</sup> )                | $J(K_v)$      | 1627.2220         | 6092.6950      |
| Moment of gear inertia per unit facewidth (kg.mm)                | $J^*$         | 36.1600           | 135.3930       |
| Relative gear mass per unit facewidth (kg/mm)                    | $m^*$         | 0.0232            | 0.0405         |
| Relative gear mass per unit facewidth (kg/mm)                    | $m_{red}$     | 0.0148            |                |
| Resonance running speed (rpm)                                    | $n_{E1}$      | 11883.7990        |                |
| Resonance ratio                                                  | $N$           | 0.0840            |                |
| Resonance ratio in main resonance ratio                          | $N_s$         | 0.8500            |                |
| Resonance range                                                  | -             | subcritical range |                |
| Dynamic factor                                                   | $K_v$         | 1.0280            |                |

|                                | Symbol        | Pair1<br>Gear1       | Pair1<br>Gear2       |
|--------------------------------|---------------|----------------------|----------------------|
| Face load factor - flank       | $K_{H\beta}$  | 1.3000               |                      |
| Face load factor - root        | $K_{F\beta}$  | 1.2500               |                      |
| Transverse load factor - flank | $K_{H\alpha}$ | 1.0000               |                      |
| Transverse load factor - root  | $K_{F\alpha}$ | 1.0000               |                      |
| Number of load cycles          | $N_L$         | $6.0000 \times 10^8$ | $4.0980 \times 10^8$ |

### 3.2. Pitting

S-N curve (Pitting), Load case 1, Meshing pair: Gear1 - Gear2



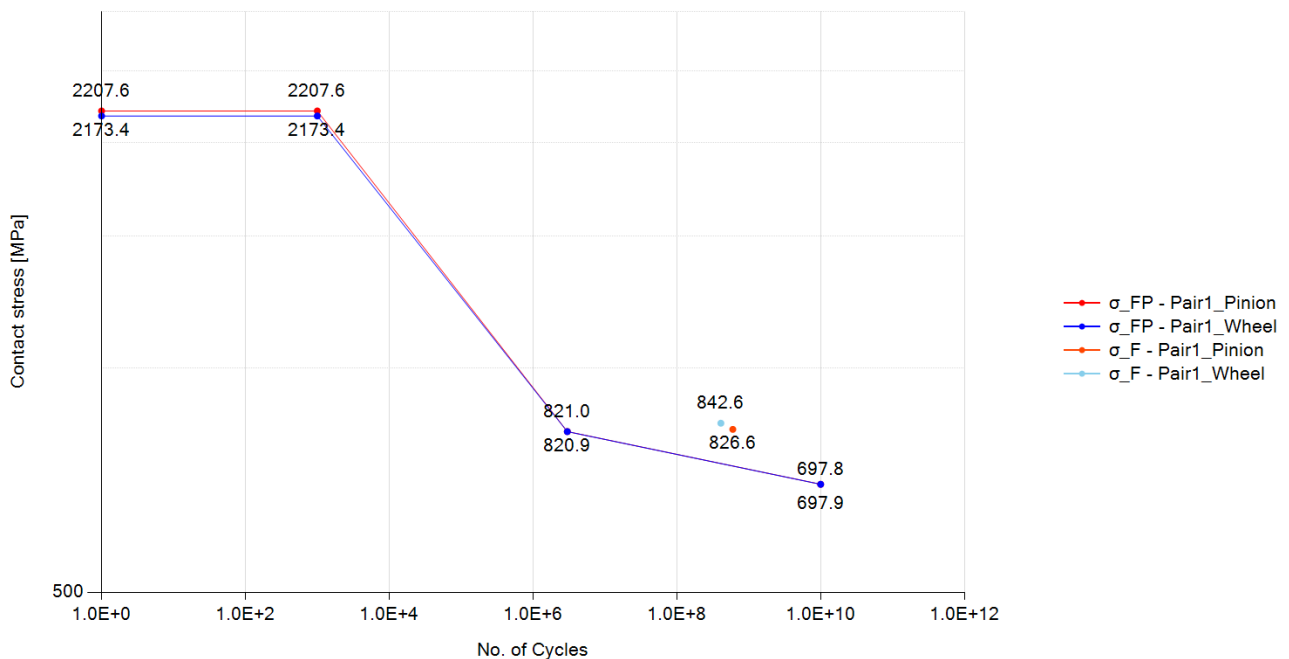
Load case 1, Meshing pair: Gear1 - Gear2

|                                                                | Symbol        | Pair1<br>Gear1       | Pair1<br>Gear2       |
|----------------------------------------------------------------|---------------|----------------------|----------------------|
| Rating Method                                                  | -             | ISO 6336-2:2019      |                      |
| Contact ratio factor                                           | $Z_\epsilon$  | 0.9070               |                      |
| Zone factor                                                    | $Z_H$         | 2.3590               |                      |
| Elasticity factor                                              | $Z_E$         | 189.8120             |                      |
| Young's modulus for tooth flank (N/mm <sup>2</sup> )           | E             | $2.0600 \times 10^5$ | $2.0600 \times 10^5$ |
| Helix angle factor                                             | $Z_\beta$     | 1.0000               |                      |
| Single pair tooth contact factor (pinion)                      | $Z_B$         | 1.0200               |                      |
| Single pair tooth contact factor (wheel)                       | $Z_D$         | 1.0000               |                      |
| Nominal contact stress at the pitch point (N/mm <sup>2</sup> ) | $\sigma_{H0}$ | 1338.1470            |                      |
| Contact stress (N/mm <sup>2</sup> )                            | $\sigma_H$    | 1763.7470            | 1729.6140            |
| Life factor                                                    | $Z_{NT}$      | 0.9270               | 0.9380               |

|                                                             | Symbol                | Pair1<br>Gear1       | Pair1<br>Gear2       |
|-------------------------------------------------------------|-----------------------|----------------------|----------------------|
| Lubricant factor                                            | Z_L                   | 0.9460               | 0.9460               |
| Velocity factor                                             | Z_V                   | 0.9800               | 0.9800               |
| Roughness factor                                            | Z_R                   | 0.9620               | 0.9620               |
| Work hardening factor                                       | Z_W                   | 1.0000               | 1.0000               |
| Size factor                                                 | Z_X                   | 1.0000               | 1.0000               |
| Permissible contact stress - static (N/mm <sup>2</sup> )    | $\sigma_{HP\_static}$ | 2400.0000            | 2400.0000            |
| Permissible contact stress - mid point (N/mm <sup>2</sup> ) | $\sigma_{HP\_mid}$    | 0.0000               | 0.0000               |
| Permissible contact stress - reference (N/mm <sup>2</sup> ) | $\sigma_{HP\_ref}$    | 1336.7950            | 1336.7950            |
| Permissible contact stress - long (N/mm <sup>2</sup> )      | $\sigma_{HP\_long}$   | 1136.2760            | 1136.2760            |
| Permissible contact stress (N/mm <sup>2</sup> )             | $\sigma_{HP}$         | 1238.6900            | 1253.2650            |
| Safety factor for Contact                                   | S_H                   | 0.7020               | 0.7250               |
| Contact damage (%)                                          | -                     | 22776.8430           | 12639.7040           |
| Cycle to failure                                            | -                     | $2.6340 \times 10^6$ | $3.2420 \times 10^6$ |

### 3.3. Bending

#### S-N curve (Bending), Load case 1, Meshing pair: Gear1 - Gear2



#### Load case 1, Meshing pair: Gear1 - Gear2

|                          | Symbol | Pair1<br>Gear1  | Pair1<br>Gear2 |
|--------------------------|--------|-----------------|----------------|
| Rating Method            | -      | ISO 6336-3:2019 |                |
| Tooth form factor        | Y_F    | 1.3820          | 1.4340         |
| Stress correction factor | Y_S    | 2.0610          | 2.0250         |

|                                                                | Symbol                | Pair1<br>Gear1       | Pair1<br>Gear2       |
|----------------------------------------------------------------|-----------------------|----------------------|----------------------|
| Working angle (deg)                                            | $\alpha_{Fen}$        | 22.5290              | 21.6820              |
| Bending moment arm (mm)                                        | $h_{Fe}$              | 3.2800               | 3.4700               |
| Tooth thickness at root (mm)                                   | $s_{Fn}$              | 6.4800               | 6.5630               |
| Tooth root radius (mm)                                         | $\rho_F$              | 1.4220               | 1.4500               |
| Helix angle factor                                             | $Y_{\beta}$           | 1.0000               |                      |
| Deep tooth factor                                              | $Y_{DT}$              | 1.0000               |                      |
| Rim thickness factor                                           | $Y_B$                 | 1.0000               | 1.0000               |
| Nominal tooth root stress (N/mm <sup>2</sup> )                 | $\sigma_{F0}$         | 514.5480             | 524.4740             |
| Tooth root stress (N/mm <sup>2</sup> )                         | $\sigma_F$            | 826.6410             | 842.5880             |
| Life factor                                                    | $Y_{NT}$              | 0.8990               | 0.9060               |
| Relative notch sensitive factor                                | $Y_{\delta relT}$     | 0.9980               | 0.9980               |
| Surface factor                                                 | $Y_{RrelT}$           | 0.9570               | 0.9570               |
| Size factor                                                    | $Y_X$                 | 1.0000               | 1.0000               |
| Stress correction factor                                       | $Y_{ST}$              | 2.0000               |                      |
| Mean stress influence factor                                   | $Y_M$                 | 1.0000               | 1.0000               |
| Technical factor                                               | $Y_T$                 | 1.0000               | 1.0000               |
| Permissible tooth root stress - static (N/mm <sup>2</sup> )    | $\sigma_{FP\_static}$ | 2207.6110            | 2173.4440            |
| Permissible tooth root stress - reference (N/mm <sup>2</sup> ) | $\sigma_{FP\_ref}$    | 821.0430             | 820.9200             |
| Permissible tooth root stress - long (N/mm <sup>2</sup> )      | $\sigma_{FP\_long}$   | 697.8860             | 697.7820             |
| Permissible tooth root stress (N/mm <sup>2</sup> )             | $\sigma_{FP}$         | 738.3540             | 743.9060             |
| Safety factor for Bending                                      | $S_F$                 | 0.8930               | 0.8830               |
| Bending damage (%)                                             | -                     | $3.2194 \times 10^5$ | $2.6920 \times 10^5$ |
| Cycle to failure                                               | -                     | $1.8640 \times 10^5$ | $1.5220 \times 10^5$ |

### 3.4. Scuffing

Load case 1, Meshing pair: Gear1 - Gear2

|                                       | Symbol         | Pair1<br>Gear1    | Pair1<br>Gear2 |
|---------------------------------------|----------------|-------------------|----------------|
| Lubrication type                      | -              | Spray lubrication |                |
| Lubricant temperature (deg)           | $\theta_{oil}$ | 80.0000           |                |
| Lubricant viscosity (mPa.s)           | $\eta_{oil}$   | 13.1510           |                |
| Load stage according to FZG A/8, 3/90 | $S_{FZG}$      | 12.0000           |                |
| Gear density (kg/m <sup>3</sup> )     | $\rho_M$       | 7830.0000         | 7830.0000      |
| Gear specific heat (J/kg.K)           | $c_M$          | 485.0000          | 485.0000       |
| Gear heat conductivity (N/s.K)        | $\lambda_M$    | 50.0000           | 50.0000        |
| Gear roughness ( $\mu m$ )            | $R_a$          | 0.6000            | 0.6000         |
| Number of mating gears                | $n_p$          | 1.0000            |                |
| Effective facewidth (mm)              | $b_{eff}$      | 38.2500           |                |

|                                                                       | Symbol    | Pair1<br>Gear1 | Pair1<br>Gear2 |
|-----------------------------------------------------------------------|-----------|----------------|----------------|
| Relevant tip relief ( $\mu\text{m}$ ) (Min. value = 2 $\mu\text{m}$ ) | C_a       | 2.0000         | 2.0000         |
| Transverse unit load (N/mm)                                           | w_Bt      | 905.5120       |                |
| Lubrication system factor                                             | X_S       | 1.2000         |                |
| Multiple meshing factor                                               | X_mp      | 1.0000         |                |
| Structural factor                                                     | X_W       | 1.0000         | 1.0000         |
| Structural factor (total)                                             | X_Wtot    | 1.0000         |                |
| Thermal contact factor                                                | B_M       | 13.7800        | 13.7800        |
| Multiple path factor                                                  | K_mp      | 1.0000         |                |
| Optimal tip relief ( $\mu\text{m}$ )                                  | C_eff     | 52.9250        |                |
| Angle factor                                                          | X_αβ      | 1.0090         |                |
| Lubricant factor                                                      | X_L       | 0.8790         |                |
| Roughness factor                                                      | X_R       | 0.7750         |                |
| Mean coefficient of friction                                          | μ_m       | 0.0800         |                |
| Maximum flash temperature (deg)                                       | θ_fl_max  | 57.3660        |                |
| Mean flash temperature (deg)                                          | θ_fl_mean | 35.9370        |                |
| Total mass temperature (deg)                                          | θ_M       | 100.2690       |                |
| Scuffing temperature (deg)                                            | θ_S       | 364.8370       |                |
| Maximum contact temperature (deg)                                     | θ_Bmax    | 157.6350       |                |
| Safety factor for flash temperature                                   | S_B       | 3.6690         |                |

### 3.5. Micro-pitting

Load case 1

|                                                                  | Symbol  | Gear1 - Gear2          |
|------------------------------------------------------------------|---------|------------------------|
| minimum lubricant film thickness ( $\mu\text{m}$ )               | h_Ymin  | 0.0513                 |
| minimum specific lubricant film thickness in the contact area    | λ_GFmin | 0.0855                 |
| permissible specific lubricant film thickness                    | λ_GFP   | 0.2111                 |
| safety factor against micropitting                               | S_λ     | 0.4053                 |
| transverse tangential load at reference cylinder per mesh (N)    | F_t     | 24390.2440             |
| transverse load in plane of action (N)                           | F_bt    | 25955.5550             |
| reduced modulus of elasticity (N/mm <sup>2</sup> )               | E_r     | 2.2637×10 <sup>5</sup> |
| thermal contact coefficient of pinion (N/(ms <sup>0.5</sup> *K)) | B_M1    | 13779.6040             |
| thermal contact coefficient of wheel (N/(ms <sup>0.5</sup> *K))  | B_M2    | 13779.6040             |
| failure load stage according to FVA 54/7 (90°)                   | -       | 0.0000                 |
| effective arithmetic mean roughness value ( $\mu\text{m}$ )      | R_a     | 0.6000                 |
| roughness factor                                                 | X_R     | 1.1020                 |
| lubricant factor                                                 | X_L     | 1.0000                 |
| lubrication factor                                               | X_S     | 1.2000                 |



|                                                                                                          | Symbol                | Gear1 - Gear2            |
|----------------------------------------------------------------------------------------------------------|-----------------------|--------------------------|
| elasticity factor (N/mm <sup>2</sup> ) <sup>0.5</sup>                                                    | Z_E                   | 189.8120                 |
| helical load factor                                                                                      | K_B $\gamma$          | 1.0000                   |
| load losses factor                                                                                       | H_v                   | 0.1220                   |
| mean coefficient of friction                                                                             | $\mu_m$               | 0.0860                   |
| profile position at min. lubricant film thickness                                                        | -                     | 0.0000                   |
| tip relief factor                                                                                        | X_Ca                  | 1.0000                   |
| load sharing factor                                                                                      | X_A                   | 0.3330                   |
| buttressing factor                                                                                       | X_but,A               | 0.0000                   |
| effective tip relief ( $\mu\text{m}$ )                                                                   | C_eff                 | 0.0000                   |
| Applied tip relief in calculation ( $\mu\text{m}$ )                                                      | C_a                   | 0.0000                   |
| sum of tangential velocities at point A (m/s)                                                            | $v_{\Sigma,A}$        | 3.1530                   |
| sliding velocity at point A (m/s)                                                                        | $v_{g,A}$             | -1.1140                  |
| tangential velocity on pinion at point A (m/s)                                                           | $v_{r1,A}$            | 1.0190                   |
| tangential velocity on wheel at point A (m/s)                                                            | $v_{r2,A}$            | 2.1340                   |
| oil inlet/sump temperature (°C)                                                                          | $\theta_{oil}$        | 80.0000                  |
| bulk temperature (°C)                                                                                    | $\theta_M$            | 98.3040                  |
| flash temperature at point A (°C)                                                                        | $\theta_{fl,A}$       | 56.7340                  |
| contact temperature at point A (°C)                                                                      | $\theta_{B,A}$        | 155.0380                 |
| pressure-viscosity coefficient at 38 °C (m <sup>2</sup> /N)                                              | $\alpha_{38}$         | $1.8390 \times 10^{-8}$  |
| pressure-viscosity coefficient at bulk temperature (m <sup>2</sup> /N)                                   | $\alpha_{\theta M}$   | $1.3440 \times 10^{-8}$  |
| pressure-viscosity coefficient at point A contact temperature (m <sup>2</sup> /N)                        | $\alpha_{\theta B,A}$ | $1.0050 \times 10^{-8}$  |
| kinematic viscosity at 40 °C (mm <sup>2</sup> /s)                                                        | $\nu_{40}$            | 68.0000                  |
| kinematic viscosity at 100 °C (mm <sup>2</sup> /s)                                                       | $\nu_{100}$           | 8.8000                   |
| kinematic viscosity at point A contact temperature (mm <sup>2</sup> /s)                                  | $\nu_{\theta B,A}$    | 3.2450                   |
| dynamic viscosity at 38 °C (Ns/m <sup>2</sup> )                                                          | $\eta_{38}$           | 0.0650                   |
| dynamic viscosity at bulk temperature (Ns/m <sup>2</sup> )                                               | $\eta_{\theta M}$     | 0.0080                   |
| dynamic viscosity at oil inlet/sump temperature (Ns/m <sup>2</sup> )                                     | $\eta_{\theta oil}$   | 0.0120                   |
| dynamic viscosity at point A contact temperature (Ns/m <sup>2</sup> )                                    | $\eta_{\theta B,A}$   | 0.0030                   |
| normal radius of relative curvature at point A (mm)                                                      | $\rho_{n,A}$          | 7.3400                   |
| material parameter                                                                                       | G_M                   | 3041.4620                |
| local velocity parameter at point A                                                                      | U_A                   | $7.1850 \times 10^{-12}$ |
| local load parameter at point A                                                                          | W_A                   | 0.0002                   |
| local sliding parameter at point A                                                                       | S_GF,A                | 0.2520                   |
| local nominal Hertzian contact stress at point A (Method B)<br>(N/mm <sup>2</sup> )                      | $p_{H,A}$             | 971.4730                 |
| local Hertzian contact stress at point A including the load factors K<br>(Method B) (N/mm <sup>2</sup> ) | $p_{dyn,A}$           | 1255.6720                |

## 4. Power loss

### 4.1. Mesh loss

Load case 1

|                                                     | Symbol           | Gear1 - Gear2       |
|-----------------------------------------------------|------------------|---------------------|
| Rating Method                                       | -                | ISO/TR 14179-2:2001 |
| Input Power (kW)                                    | P                | 107.2739            |
| Gear ratio                                          | u                | 1.4643              |
| Tangential speed at working pitch dia. (m/s)        | v <sub>wt</sub>  | 4.4620              |
| Tangential force at pitch dia. (N)                  | F <sub>t</sub>   | 24390.2439          |
| Tooth normal force, transverse section (N)          | F <sub>bt</sub>  | 25955.5554          |
| Arithmetic average roughness of pinion              | Ra1              | 0.6000              |
| Arithmetic average roughness of wheel               | Ra2              | 0.6000              |
| Roughness factor                                    | X <sub>R</sub>   | 0.6000              |
| Sum velocity (m/s)                                  | v <sub>Σ</sub>   | 3.3631              |
| Equivalent radius of curvature (mm)                 | ρ                | 9.5415              |
| Kinematic viscosity of oil at operating temp. (cSt) | η <sub>oil</sub> | 14.8602             |
| Lubricant type                                      | -                | Mineral oil base    |
| Lubricant factor                                    | X <sub>L</sub>   | 1.0000              |
| distributed force on tooth surface(N/m)             | F/b              | 576.7901            |
| Mean coefficient of friction of the gear mesh       | μ <sub>mz</sub>  | 0.0658              |
| Tooth loss factor                                   | H <sub>v</sub>   | 0.1223              |
| Mesh power loss (W)                                 | P <sub>VZP</sub> | 863.2802            |
| Mesh efficiency (%)                                 | η <sub>z</sub>   | 99.1953             |

### 4.2. Churning loss

Load case 1

|                                              | Symbol         | Gear1            | Gear2            |
|----------------------------------------------|----------------|------------------|------------------|
| Rating Method                                | -              | Changenet (2011) | Changenet (2011) |
| Rotating speed (rpm)                         | n              | 1000.0000        | 682.9268         |
| Lubricant                                    | -              | Oil: ISO-VG 68   | Oil: ISO-VG 68   |
| Gear immersed depth (mm)                     | h <sub>e</sub> | 42.9396          | 62.1386          |
| Oil density (kg/m <sup>3</sup> )             | ρ              | 885.0000         | 885.0000         |
| Operating Viscosity (cSt)                    | ν              | 14.8602          | 14.8602          |
| dynamic viscosity (kg/m.s)                   | μ              | 0.0000           | 0.0000           |
| Reynolds number                              | Re             | 13616.7852       | 13457.1189       |
| Froude number                                | Fr             | 48.0006          | 32.3965          |
| Centrifugal acceleration (m/s <sup>2</sup> ) | γ              | 196.9947         | 103.9212         |

|                                  | Symbol | Gear1   | Gear2    |
|----------------------------------|--------|---------|----------|
| Tip diameter of gear (mm)        | d_a    | 91.8792 | 130.2772 |
| Pitch diameter of gear (mm)      | d_p    | 85.8792 | 124.2772 |
| Facewidth (mm)                   | b      | 45.0000 | 45.0000  |
| Dimensionless drag torque        | C_m    | 0.0213  | 0.0223   |
| Immersion area of the gear (m^2) | S_m    | 0.0209  | 0.0342   |
| Total oil churning loss (W)      | P      | 17.9142 | 29.6216  |
| Total oil churning torque (N.m)  | T      | 0.1711  | 0.4142   |

### 4.3. Windage loss

Load case 1

|                      | Symbol | Gear1  | Gear2  |
|----------------------|--------|--------|--------|
| Rating Method        | -      | None   | None   |
| Windage loss (W)     | P_WL   | 0.0000 | 0.0000 |
| Windage torque (N.m) | T_WL   | 0.0000 | 0.0000 |